Appendix A. Contracts, features and account checks

Data and trade construction

WRDS supplies option quotes, stock prices, rates and distributions. The inherited formation archive contains 345 pairs; 125 prepared pairs with 161 equity/ETF underlyings supply candidates. Forty-six pairs generate 608 screened put proposals over 2018–2025. This universe is retrospective and conditioned on usable coverage. The companion roster in paper/pair_universe.csv records prepared pairs and candidate-date ranges, which are not claims of continuous quote availability.

Put selection takes the expiry nearest 30 days, then the put nearest absolute delta 0.25 within 0.075. Ties prefer narrower quotes, higher open interest and lower contract identifier. Entry requires standard 100-share contracts, positive noncrossed quotes, open interest of at least 100, quote width no more than 25% of midpoint and 7–55 days remaining. Straddles match call and put on name, strike, expiry and adjustment. Corporate events can force exits; missing entry quotes reject proposals. Held marks may use last-known IV with current spot, rates and known dividends.

Entry-state definitions

Put-pair feature group	Numerical inputs
Volatility and opportunity	Mean IV; oriented IV gap; rank strength; oriented IV²-minus-trailing-variance gap; mean RV
Relatedness and maturity	126-session beta and correlation; mean expiry and expiry difference
Costs and market	Training commission in IV points; strictly prior VIX close
Completed profitability	91-day and 365-day mean profit per initial vega and completed counts

These are 15 numerical inputs. The first three are core fields; the remaining 12 have binary availability flags, giving 27 inputs. Missing optional values are zero with flag zero. Put trailing variance is 252 times mean squared simple stock returns over 20 sessions, with at least 10 observations. The straddle signal uses annualized squared log returns and a 21-session RV input. The measures have different retained definitions; no new common RV estimator is fitted for this comparison.

The straddle's 14 ridge inputs are completed residual memory, IV, VIX, normalized memory, variance ratio, 21-session VIX change, 5-session IV change, and a fixed-half benchmark's 21/63/126-session Sharpe, 63-session volatility and drawdown, reserve use and availability flag. These benchmark states come from an external causal paper portfolio, not the learner's own future inventory.

Admission and risk accounting

Research stress takes each name's worst modeled option-plus-hedge loss across spot shocks −20/0/+20% and IV shocks −10/0/+10 points, then sums without pair-correlation credit. Limits are stress loss 5% of budget, reserve 80%, hedge notional twice budget and gross vega $20,000 per IV point per $1 million. Reserve is gross option premium plus half netted hedge notional. Global checks use equity; pair checks use a $1 million reference scaled by current/initial equity. These are research limits, not broker margin. European-style valuation and cash settlement approximate expiry; assignment is not fully simulated.

Appendix B. Fitting, chronology and controls

Entry optimizers

The put tree has up to eight leaves, with minimum count max(8, min(63, floor(n/8))); fewer than 16 examples use full size. CEM uses 50 rounds of 192 maps, 19 elites and seed 2026. Probabilities start uniform; updates mix 30% prior probability and 70% elite frequency, shrink by 0.97 and add 0.03 uniformly. Constant maps and the incumbent are anchors. Separate rounded-contract paths supply daily training profits; funded NAV returns supply evaluation Sharpe. Put memory and training labels retain a $0.65 option commission; funded evaluation uses $0.03 for every policy.

The straddle ridge fit uses median centers, standard deviations floored at 0.001, clipped inputs within ±5, penalty 100 with no intercept penalty, and equal total weight per date. Full-size completed profit is its target. Score terciles define contexts; all 27 size maps are evaluated on aggregate daily paths. Ties favor lower exposure. The 60-month window requires 128 examples. Only 4,116 training rewards are finite; the policy still handles all 35,774 proposals.

Q specification and numerical check

$$Q(s, a) = b(s) + a\,c(s) + a^2 d(s), \qquad a \in \{0, .10, .25, .50, .75, .90, 1\}$$

Each coefficient function is linear in the intercept and nine standardized inputs. Training centers use means, scales have floor 0.1, and standardized values are clipped within ±5. Ridge is 10, except the global intercept penalty 0.01. Bellman regression weights counterfactual rows so each week has unit total mass. The reference volatility is that family's completed-window 50/50 daily excess standard deviation. All 81 monthly fits were checked at 200, 500, 1,000 and 2,000 updates. Later actions are identical across budgets for the four reported λ values.

Allocation control, λ = .20	2018–21 S	2022–25 S	Later profit
Fixed at Q's mean target	0.828	1.387	$0.841m
One-period learner, γ = 0	1.053	1.645	$1.184m
Sequential Q, γ = 0.99	1.215	1.726	$1.306m

The mean-weight control uses the realized average target retrospectively. Forty-nine nonzero circular rotations of the saved λ = 0.20 action schedule preserve its action counts and most serial structure; none reaches actual later Sharpe, while three reach or exceed actual validation Sharpe. Their median later Sharpe is 0.744. Wealth and resizing costs are recomputed. These are descriptive timing diagnostics, not selection-adjusted tests.

Chronology and statistical interpretation

Every fit uses strictly completed past outcomes. Put selection ranks 33 archived screened configurations by 2018–2021 scores before the $10 million replay. Straddle specification search inspected both reporting periods; its record is preserved. The four Q preferences describe researched risk choices and do not replace the original validation-selection record. Chronological fitting and selection independence are distinct. The put interval uses 1,000 paired circular resamples of 126-session blocks, with fixed policies and no research-selection adjustment. A prospectively frozen evaluation remains open.

Appendix C. Supporting results and reproducibility

Year	Put bandit	Straddle bandit	Q: λ = .20	Q: λ = 1
2018	0.522	0.148	0.160	0.160
2019	-0.542	1.459	0.112	-0.237
2020	1.555	0.161	1.617	1.028
2021	-0.580	2.336	2.932	3.180
2022	2.458	1.206	2.042	2.540
2023	1.492	1.727	2.449	2.907
2024	-0.907	1.465	1.945	1.954
2025*	1.092	0.085	-0.598	1.092

All figures are annualized excess Sharpe; 2025 ends on 29 August. The sleeves retain their separate funded histories, and outer allocations include resizing charges. Annual results show regime variation that aggregate ratios conceal. A high pooled Sharpe is not evidence of positive excess performance every year.

Entry policy + Q	λ = 0	λ = .10	λ = .20	λ = 1
Fixed 50%	0.541	0.703	1.116	1.349
Fixed 75%	0.606	0.667	2.118	1.897
Fixed 100%	0.618	1.354	1.194	1.633
Bandit	0.671	1.491	1.726	2.152

This table holds Q's feature definitions, discount, training window and action menu fixed while refitting separately on each entry family. Values are 2022–2025 Sharpes. The complete seven fixed capital splits are also retained; 50/50 is the primary fixed-allocation baseline, rather than a later-period optimized control.

Evidence and reproduction

Entry results come from reports/p25_validation_split/periods.csv and reports/straddle_policy_refinement/comparison.csv. Exact entry explanations are in reports/retained_entry_explanations. Allocation results, decisions and convergence checks are in reports/signal_allocation_variance. Independent fixed-entry Q fits and the four-way experiment are in reports/signal_learning_factorial. Its checks reproduce all eight pure strategy endpoints and 324 monthly checkpoint timings. The paper's provenance manifest records source and figure hashes.

The companion scripts preserve the actual-contract ledger and the separate fractional-index allocator. Model fitting requires the licensed source cache or archived derived inputs; vendor redistribution restrictions still apply. Reproduce this document from its saved manuscript and figures without refitting any model. Public repository: https://github.com/kreitzz/rl-atm-vega-pairs

Appendix D. Original sizing and annual results

Alternative put sizing pipelines retained from the original study

Put sizing policy	2018–21 S	2022–25 S	Later entries	Later profit
Constant 50%	0.437	1.007	151	$0.205m
Constant 75%	0.449	0.944	104	$0.195m
Constant 100%	0.478	1.055	75	$0.201m
Tree + CEM	0.573	1.263	110	$0.249m
Spread-rank sizing	0.464	0.815	121	$0.165m
Ridge-score sizing	0.421	0.890	94	$0.180m
Direct linear policy	0.267	0.873	96	$0.171m
Direct additive policy	0.302	1.013	97	$0.180m

All policies receive the same put proposals, directions, contracts and funded admission rules. The lower four rows are follow-up comparisons rather than original selection candidates. Tree/CEM has the highest funded Sharpe in both periods among these policies. These comparisons retain the original question about entry sizing alongside the newer straddle and allocation experiments.

Year	Account return	Above cash (pp)	Excess Sharpe	Entries
2022	3.470%	1.673	2.458	66
2023	5.288%	0.432	1.492	19
2024	5.104%	-0.283	-0.907	11
2025*	3.207%	0.169	1.092	14

Calendar-period account returns and above-cash differences are not annualized, including partial 2025. Sharpe is annualized. In 2024 the account grows while losing relative to cash. The original put strategy's results are concentrated in a few pairs and in 2022; this limits claims of broad repeated success.

The original put results are concentrated: SNAP/UBER, GAP/TJX and AMD/MU contribute 66.9% of later net pair trading profit, and 2022 supplies 83.1%. These figures exclude shared financing. They describe where gains occurred, not a rule for selecting those names retrospectively. Annual Sharpe ratios are retained in Appendix C.

The put account's later annualized total account return is 4.667%, versus 0.530% growth relative to cash. Average reserve is 2.44% and peak reserve 15.77%. Its cash-relative drawdown is 0.495%, compared with 0.646% at full size. These figures retain the original distinction between a profitable trading signal, account funding and capital deployment.

Appendix E. Performance definitions and timing diagnostics

Account return, cash-relative growth and dollar profit

Let N be account NAV and K the continuing all-cash benchmark. Both begin at $10 million in January 2018. The benchmark compounds the previous available WRDS zero-curve rate nearest 30 days over calendar days using a 365-day year. For each reporting period, 0 is the preceding close and T the final close.

$$e_t = \frac{N_t}{N_{t-1}} - \frac{K_t}{K_{t-1}}, \qquad \widehat{S} = \sqrt{252}\,\overline{e}/s_e.$$

Dollar profit above cash is (N_T − K_T) − (N_0 − K_0). With Q_t = N_t/K_t and Y equal to elapsed days divided by 365.25, annualized growth relative to cash is (Q_T/Q_0)^(1/Y) − 1. Cash-relative drawdown is the greatest percentage decline of Q from its running peak, including the opening close. The annual table instead subtracts the cash return from the account return, in percentage points.

At the 2021 close, the learned put account has $10.568 million equity and four open trades; full sizing has $10.581 million and three. Cash benchmark wealth is $10.510 million. Performance curves subtract each policy's opening dollar excess wealth; they do not reset accounts or liquidate inventory at the reporting split.

Additional allocation diagnostics

Centering VIX and straddle IV-versus-RV richness changes 13.6% and 12.6% of later actions. Across 49 nonzero rotations of Q's saved schedule, none reaches its later Sharpe; median Sharpe is 0.744. These diagnostics explain information use and timing, without establishing causal feature effects or selection-adjusted significance.

The rotation control preserves allocation counts and most serial structure, then recomputes wealth and resizing costs. Three of 49 rotated schedules reach or exceed the actual validation Sharpe; none reaches the actual later Sharpe. Timing-control results are reported in Appendix B. These are diagnostics of the retained policy, not a substitute for prospective evaluation.

A final distinction concerns portfolio admission. The source put and straddle books each simulate whole contracts and daily stock hedges under their own funded histories. The outer allocator trades fractional claims on those net strategy indices, with additional resizing charges. Its capital weights are not premium deployment or broker margin, and reallocating indices does not rerun a common integer-contract inventory ledger.